Central simple algebras in number theory

An explicit tour: quaternion and cyclic algebras, the local invariant, reciprocity, and the theorems they prove

cohomological readings are set off in grey boxes and may be skipped; computations in `csa-brauer.gp`

How to read this. The argument runs on explicit algebras — quaternion algebras (a, b), cyclic algebras $(L/K, \sigma, b)$, Hilbert symbols — and never needs class field theory as an input. On the contrary: the Brauer group is one of the standard ways to **state and prove** class field theory, and that is how it appears below. Wherever an explicit step is a cohomological statement in disguise, a grey box says which.

1. The objects

Let K be a field. A **central simple algebra** (CSA) over K is a finite-dimensional K-algebra A with centre exactly K and no two-sided ideals but 0 and A.

Wedderburn. Every CSA over K is $A \cong M_m(D)$ for a division algebra D with centre K, and D is determined by A up to isomorphism. Moreover $\dim_K A$ is always a square, say $\dim_K A = n^2$; n is the **degree**, and $\sqrt{\dim_K D}$ is the **index** of A.

Two CSAs are **Brauer equivalent** if they have the same D. The set of classes, with the product induced by $\otimes_K$, is a group: the **Brauer group** $\mathrm{Br}(K)$. The identity is the class of $M_n(K)$; the inverse of A is the opposite algebra A^{op}, because $A \otimes_K A^{\mathrm{op}} \cong M_{n^2}(K)$. A field $L \supseteq K$ **splits** A if $A \otimes_K L \cong M_n(L)$; the classes split by L form a subgroup $\mathrm{Br}(L/K)$.

Everything below is built from two families of examples, and they are completely explicit.

1.1. Quaternion algebras

For $a, b \in K^\times$ (char $K \neq 2$), let

$$(a, b)_K = K\langle i, j\rangle \quad \text{with} \quad i^2 = a, \quad j^2 = b, \quad ij = -ji.$$

This is a CSA of degree 2, with K-basis $1, i, j, ij$. There are only two possibilities: it is a division algebra, or it is $M_2(K)$ — in which case one says it **splits**.

Proposition 1.1. The following are equivalent:

(i) $(a, b)_K$ splits; (ii) the conic $ax^2 + by^2 = z^2$ has a K-point other than the origin; (iii) b is a norm from $K(\sqrt{a})$; (iv) the quadratic form $\langle a, b, -1\rangle$ represents 0.

Sketch. The reduced norm of $x + yi + zj + wij$ is $x^2 - ay^2 - bz^2 + abw^2$. An algebra of degree 2 is split iff its norm form is isotropic, and one checks directly that the norm form is isotropic iff $\langle a, b, -1\rangle$ is. For (iii): (a, b) contains $K(\sqrt{a}) = K(i)$, and $jxj^{-1} = \overline{x}$ for $x \in K(i)$, so (a, b) is the cyclic algebra of $K($`sqrt` $a)/K$ with twisting element b; a degree-2 cyclic algebra splits iff the twisting element is a norm. ■

The Hilbert symbol is the indicator of this: for a local field K_v put

$$(a, b)_v = \begin{cases} +1 \text{ if } (a, b)_{K_v} \text{ splits} \\ -1 \text{ otherwise.} \end{cases}$$

This is the single most useful explicit object in the subject, and PARI computes it as `hilbert(a,b,p)` (with `p = 0` for the real place).

1.2. Cyclic algebras

Let L/K be cyclic of degree n with $\mathrm{Gal}(L/K) = \langle\sigma\rangle$, and $b \in K^\times$. Put

$$(L/K, \sigma, b) = \oplus_{k=0}^{n-1} L u^k, \quad u^n = b, \quad ux = \sigma(x)u \ \text{ for } \ x \in L.$$

This is a CSA of degree n over K, split by L.

Proposition 1.2. $(L/K, \sigma, b)$ splits if and only if $b \in N_{L/K}(L^\times)$. More generally $(L/K, \sigma, b) \cong (L/K, \sigma, b')$ iff $b/b' \in N_{L/K}(L^\times)$, so

$$b \mapsto (L/K, \sigma, b) \quad \text{induces} \quad K^\times / N_{L/K}(L^\times) \cong \mathrm{Br}(L/K).$$

This one isomorphism is the engine of everything that follows: **it converts a norm question into an algebra question.** The quaternion algebra (a, b) is the case $L = K(\texttt{sqrt}\ a)$, $n = 2$.

Cohomological reading. $\mathrm{Br}(K) \cong H^2\big(G_K, \overline{K}^\times\big)$, with $\mathrm{Br}(L/K) = H^2(\mathrm{Gal}(L/K), L^\times)$; the cyclic algebras realise the periodicity of the cohomology of a cyclic group, $H^2(\mathbb{Z}/n, L^\times) = K^\times/N(L^\times)$, which is Proposition 1.2. The quaternion algebra (a, b) is the cup product of the classes of a and b under $K^\times/(K^\times)^2 = H^1(K, \mu_2)$, and the Hilbert symbol is that cup product evaluated by the local invariant of Section 2. "Crossed product algebra" is the explicit name for a 2-cocycle; Section 6 makes this precise and derives it.

The **exponent** (or period) of A is its order in $\mathrm{Br}(K)$; the **index** is $\sqrt{\dim_K D}$. Always exponent | index and they have the same prime factors. Over number fields they are **equal** (Section 5.3) — a theorem with no analogue over general fields.

2. Local fields: the invariant, explicitly

Let K_v be a non-archimedean local field with residue field $\mathbb{F}_q$, uniformiser π, and let K_n be **the** unramified extension of degree n, cyclic with canonical generator the Frobenius φ.

Theorem 2.1. Every CSA over K_v is split by an unramified extension. Concretely, every class in $\mathrm{Br}(K_v)$ is $(K_n/K_v, \varphi, \pi^r)$ for some n and r, and

$$\mathrm{inv}_v(K_n/K_v, \varphi, \pi^r) = \frac{r}{n} \in \mathbb{Q}/\mathbb{Z}$$

is a well-defined isomorphism

$$\mathrm{inv}_v : \mathrm{Br}(K_v) \cong \mathbb{Q}/\mathbb{Z}.$$

The index of the class of invariant r/n (in lowest terms) is exactly n.

So the local Brauer group is $\mathbb{Q}/\mathbb{Z}$, and **the invariant is nothing but a valuation divided by a degree**. Two consequences to keep in mind:

- inv_v of a quaternion algebra is 0 or 1/2, and $(a, b)_v = (-1)^{2\mathrm{inv}_v(a,b)}$. The Hilbert symbol is the invariant, written multiplicatively.
- For the archimedean places: $\mathrm{Br}(\mathbb{R}) = \{0, \mathbb{H}\} \cong \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ with $\mathbb{H}$ Hamilton's quaternions $(-1, -1)_\mathbb{R}$, and $\mathrm{Br}(\mathbb{C}) = 0$.

Local class field theory, for free. Combining 1.2 and 2.1: for L/K_v cyclic of degree n, $\mathrm{Br}(L/K_v)$ is the subgroup $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, so

$$K_v^\times / N_{L/K_v}(L^\times) \cong \frac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathrm{Gal}(L/K_v).$$

That is the local reciprocity isomorphism, and it fell out of counting invariants.

Cohomological reading. inv_v is the composite $H^2(\mathrm{Gal}(K_n/K_v), K_n^\times) \xrightarrow{v} H^2(\mathrm{Gal}(K_n/K_v), \mathbb{Z}) \cong H^1(\mathrm{Gal}(K_n/K_v), \mathbb{Q}/\mathbb{Z}) \xrightarrow{\varphi\mapsto 1/n} \frac{1}{n}\mathbb{Z}/\mathbb{Z}$, the first map induced by the valuation (the units contribute nothing, by Hilbert 90 and the vanishing of H^2 of the unit group in the unramified case). The class of invariant $1/n$ is the **fundamental class**, and the whole of local class field theory is the statement that (K_v, inv) is a class formation.

3. The two global theorems

Now let K be a number field. A CSA A over K has a local invariant at each place, $\mathrm{inv}_v(A \otimes_K K_v) \in \mathbb{Q}/\mathbb{Z}$, zero for all but finitely many v (those where A has "good reduction"). The whole of global class field theory is packaged in one exact sequence.

Theorem 3.1 (the fundamental sequence).

$$0 \longrightarrow \mathrm{Br}(K) \longrightarrow \oplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

The two non-trivial assertions are:

(a) Albert–Brauer–Hasse–Noether. A splits over K if and only if $A \otimes_K K_v$ splits for every place v. (Injectivity: a Hasse principle for algebras.)

(b) Reciprocity. $\sum_v \mathrm{inv}_v(A) = 0$ for every A. (The composite is zero, and the sequence is exact.)

Part (b), written multiplicatively for quaternion algebras, is **Hilbert reciprocity**:

$$\prod_v (a,b)_v = 1 \quad \text{for all } a, b \in K^\times.$$

`csa-brauer.gp` verifies this on a sample; e.g. for $(a,b) = (2,3)$ over $\mathbb{Q}$ the only non-trivial symbols are $(2,3)_2 = (2,3)_3 = -1$, and the product is $+1$.

Cohomological reading. Theorem 3.1 is the statement that the idele class group is a class formation with fundamental class of invariant $1/n$; part (a) is $H^2\big(K, \overline{K}^\times\big) \hookrightarrow \oplus_v H^2\big(K_v, \overline{K_v}^\times\big)$, i.e. the vanishing of $\text{Ш}^2\big(K, \overline{K}^\times\big) = \text{Ш}^2(K, \mathbb{G}_m)$. Global class field theory — Artin reciprocity, the existence theorem — is **equivalent** to Theorem 3.1, and this is one of the standard routes to proving it.

4. Reciprocity is quadratic reciprocity

The cleanest illustration that Theorem 3.1(b) is not an abstraction. Let $p \neq q$ be odd primes and take $a = p$, $b = q$ in Hilbert reciprocity. Three explicit local formulas:

$$(p,q)_\infty = 1 \quad (p, q > 0), \qquad (p,q)_\ell = 1 \quad (\ell \nmid 2pq),$$

$$(p,q)_p = \left(\frac{q}{p}\right), \qquad (p,q)_q = \left(\frac{p}{q}\right), \qquad (p,q)_2 = (-1)^{\varepsilon(p)\varepsilon(q)}, \quad \varepsilon(m) = \frac{m-1}{2} \bmod 2.$$

So the product formula $\prod_v (p,q)_v = 1$ collapses to three factors and reads

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}},$$

which is Gauss's law of quadratic reciprocity.

The same trick with $a=-1$ and $a=2$ gives the two supplements: $(-1,q)_q = ((-1)/q)$ and $(-1,q)_2 = (-1)^{\varepsilon(q)}$ give $((-1)/q) = (-1)^{(q-1)/2}$; and $(2,q)_2 = (-1)^{\omega(q)}$ with $\omega(m) = (m^2-1)/8$ gives $(2/q) = (-1)^{(q^2-1)/8}$.

`csa-brauer.gp` checks the three local identities and the resulting law on a sample of pairs:

p	q	$(p,q)_p$	(q/p)	$(p,q)_q$	(p/q)	$(p,q)_2$	$(-1)^{\varepsilon\varepsilon}$
3	5	-1	-1	-1	-1	$+1$	$+1$
3	7	$+1$	$+1$	-1	-1	-1	-1
7	11	$+1$	$+1$	-1	-1	-1	-1
13	17	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$

Cohomological reading. Quadratic reciprocity is the degree-2 case of Artin reciprocity, and the product formula is the statement that the global fundamental class has total invariant zero. That the **hardest** classical theorem of elementary number theory is the **easiest** consequence of the sequence in Theorem 3.1 is the best advertisement the Brauer group has.

5. Applications

5.1. Conics, Legendre, Hasse–Minkowski

By Proposition 1.1, a conic $ax^2+by^2=z^2$ over K has a rational point iff $(a,b)_K$ splits. Applying ABHN:

Legendre's theorem. $ax^2+by^2=z^2$ has a non-trivial K-point if and only if it has one over every K_v — equivalently $(a,b)_v = 1$ for all v.

And since one place can be omitted (its symbol is determined by the others through reciprocity), one never has to check them all: **the Hasse principle for conics plus reciprocity is Legendre's criterion**, and the number of conditions is finite and explicit. `csa-brauer.gp`:

(a,b)	splits everywhere	rational point, height ≤ 40
$(2,3)$	no	none
$(3,5)$	no	none
$(1,1)$	yes	$(0,1,1)$
$(-1,-1)$	no	none — Hamilton's quaternions, ramified at ∞ and 2
$(2,7)$	yes	$(1,1,3)$
$(6,10)$	yes	$(1,1,4)$

The general Hasse–Minkowski theorem — a quadratic form over K in any number of variables is isotropic iff it is isotropic over every K_v — reduces to the ternary case by an induction, and the ternary case **is** the statement about quaternion algebras.

5.2. The Hasse norm theorem, and where it fails

Theorem (Hasse). Let L/K be **cyclic**. Then $x \in K^\times$ is a norm from L if and only if it is a local norm at every place.

Proof, in one line. By Proposition 1.2, x is a norm iff the cyclic algebra $(L/K, \sigma, x)$ splits, and x is a local norm at v iff that algebra splits over K_v. Now apply ABHN. ■

That is the whole proof, and it is a good example of what the Brauer group buys: **a statement about norms becomes a statement about an algebra, where the Hasse principle is a theorem.**

The hypothesis "cyclic" is essential, and the failure is instructive. Take $L = \mathbb{Q}\big(\sqrt{13}, \sqrt{17}\big)$, with $\mathrm{Gal}(L/\mathbb{Q}) = (\mathbb{Z}/2)^2$.

Every decomposition group of $L/\mathbb{Q}$ has order at most 2. The discriminant is $13^2 \cdot 17^2$, so only 13 and 17 ramify (in particular 2 is unramified). Now $(17/13) = 1$, so 13 splits in $\mathbb{Q}\big(\sqrt{17}\big)$; and $(13/17) = 1$, so 17 splits in $\mathbb{Q}\big(\sqrt{13}\big)$: each ramified prime has decomposition group of order 2. Every unramified prime has **cyclic** decomposition group, and $(\mathbb{Z}/2)^2$ has no element of order 4, so those have order ≤ 2 as well. The real places are split.

Consequence. At every v the local extension $L_w/\mathbb{Q}_v$ is trivial or quadratic, so $25 = N(5)$ is a local norm **everywhere**.

And yet it is not a global norm. Since the whole point is that no single place obstructs, the proof has to use a global input; it uses two, both elementary — the factorisation of 5 and the sign of the fundamental unit.

Proposition 5.1. $25 \notin N_{L/\mathbb{Q}}(L^\times)$ for $L = \mathbb{Q}\big(\sqrt{13}, \sqrt{17}\big)$.

Proof. Put $k = \mathbb{Q}\big(\sqrt{13}\big)$, so that $L = k\big(\sqrt{17}\big)$ and $N_{L/\mathbb{Q}} = N_{k/\mathbb{Q}} \circ N_{L/k}$. Suppose $25 = N_{L/\mathbb{Q}}(x)$ for some $x \in L^\times$, and set $y = N_{L/k}(x) \in k^\times$. Then

(i) $N_{k/\mathbb{Q}}(y) = 25$, and
(ii) y is a norm from $L = k\big(\sqrt{17}\big)$.

Step 1: (i) forces $y \equiv \pm 5$ modulo squares. Since $13 \equiv 3 \bmod 5$ is not a square mod 5, the prime 5 is **inert** in k; so the only ideal of k of norm 25 is (5), and (i) gives $(y) = (5)$, i.e. $y = 5u$ with u a unit of norm 1. The fundamental unit $\varepsilon = \big(3 + \sqrt{13}\big)/2$ has $N(\varepsilon) = (9 - 13)/4 = -1$, so a unit $\pm\varepsilon^n$ has norm $(-1)^n$ and the units of norm 1 are exactly $\pm\varepsilon^{2n} = \pm(\varepsilon^n)^2$. Hence

$$y \in \{5, -5\} \cdot (k^\times)^2.$$

Step 2: neither 5 nor -5 is a norm from $k\big(\sqrt{17}\big)$. By Proposition 1.1, (ii) says the quaternion algebra $(17, y)_k$ splits; and the Hilbert symbol depends only on y modulo squares, so by Step 1 it would follow that $(17, \pm 5)_k = 1$. Now $13 \equiv 8^2 \bmod 17$, so 17 **splits** in k; let w be a place of k above 17, so that $k_w = \mathbb{Q}_{17}$. There 17 is a uniformiser and ± 5 a unit, so the tame symbol is a Legendre symbol:

$$(17, \pm 5)_w = \left(\frac{\pm 5}{17}\right) = \left(\frac{5}{17}\right) = \left(\frac{17}{5}\right) = \left(\frac{2}{5}\right) = -1,$$

using $(-1/17) = 1$ because $17 \equiv 1 \bmod 4$, then quadratic reciprocity, then $17 \equiv 2 \bmod 5$. So $(17, \pm 5)_k$ is ramified at w and does not split — a contradiction. ■

Note that only the **easy** direction of the local–global principle was used: an algebra that fails to split at one place fails to split. No appeal to ABHN is needed to prove a negative.

Why 4 and 9 behave differently. The same argument run on 4 reaches $y \in \{2, -2\} \cdot (k^\times)^2$ (2 is inert in k since $13 \equiv 5 \bmod 8$), and there the symbol at $w \mid 17$ is $(2/17) = +1$ — $2 \equiv 6^2 \bmod 17$ — so no obstruction appears, and indeed 4 **is** a norm. For 9 the first step already fails to pin y down: 3 **splits** in k, so there are several ideals of norm 9 and correspondingly more candidates for y. The whole difference between 25 and 4 is the difference between $(5/17) = -1$ and $(2/17) = +1$.

`csa-brauer.gp` confirms each step: 5 inert and 17 split in k, $N(\varepsilon) = -1$, and the local symbols $(17, \pm 5)_w = -1$ at $w \mid 17$ while $(17, \pm 5)_w = +1$ at $w \mid 2$ and $w \mid 5$ — so 17 is the **only** obstructing place, as the proof claims.

5.2.1. The norm form, explicitly

It costs half a page to write $N_{L/\mathbb{Q}}$ down, and the shape is worth seeing. Take the $\mathbb{Q}$-basis $1, \sqrt{13}, \sqrt{17}, \sqrt{221}$ of L (a basis of the **field**; it is not an integral basis) and let

$$x = a + b\sqrt{13} + c\sqrt{17} + d\sqrt{221}.$$

The four conjugates are obtained by the four sign patterns $\left(\pm\sqrt{13}, \pm\sqrt{17}\right)$, with the sign on $\sqrt{221} = \sqrt{13}\sqrt{17}$ being the product. Group them in pairs. Writing $x = \left(a + b\sqrt{13}\right) + \left(c\sqrt{17} + d\sqrt{221}\right)$ and using $\sqrt{17} \cdot \sqrt{221} = 17\sqrt{13}$,

$$x \cdot \tau_1(x) = \left(a + b\sqrt{13}\right)^2 - \left(c\sqrt{17} + d\sqrt{221}\right)^2 = A + B\sqrt{13},$$

$$A = a^2 + 13b^2 - 17c^2 - 221d^2, \quad B = 2(ab - 17cd),$$

and $\tau_2(x)\tau_3(x)$ is its conjugate $A - B\sqrt{13}$. Multiplying:

$$\boxed{N_{L/\mathbb{Q}}(x) = \left(a^2 + 13b^2 - 17c^2 - 221d^2\right)^2 - 52\,(ab - 17cd)^2.}$$

A quartic form in four variables. Sanity checks: $N(5) = 625$, $N\left(\sqrt{13}\right) = 169$, $N\left(\sqrt{17}\right) = 289$, $N\left(\sqrt{221}\right) = 221^2$, $N\left(1 + \sqrt{13}\right) = 144$; and `csa-brauer.gp` checks the formula against PARI on 300 random quadruples.

Three presentations, one per quadratic subfield. Since $N_{L/\mathbb{Q}} = N_{k_i/\mathbb{Q}} \circ N_{L/k_i}$ for each of the three quadratic subfields, the **same** quartic form has three different "$A^2 - dB^2$" shapes:

σ	fixes	A_σ	$N = A_\sigma^2 - dB_\sigma^2$
τ_1	$\sqrt{13}$	$a^2 + 13b^2 - 17c^2 - 221d^2$	$d = 13,\ B = 2(ab - 17cd)$
τ_2	$\sqrt{17}$	$a^2 - 13b^2 + 17c^2 - 221d^2$	$d = 17,\ B = 2(ac - 13bd)$
τ_3	$\sqrt{221}$	$a^2 - 13b^2 - 17c^2 + 221d^2$	$d = 221,\ B = 2(ad - bc)$

The sign patterns are the character table. Each basis element $e \in \left\{1, \sqrt{13}, \sqrt{17}, \sqrt{221}\right\}$ is an eigenvector of every $\sigma \in G$, with eigenvalue ± 1, and

$$A_\sigma = \sum_e \varepsilon_\sigma(e)\, a_e^2\, e^2, \quad e^2 = 1, 13, 17, 221 \ \text{ respectively.}$$

So the three sign rows $(+,+,-,-)$, $(+,-,+,-)$, $(+,-,-,+)$ are exactly the three non-trivial characters of $(\mathbb{Z}/2)^2$. The cross term B_σ pairs the two basis elements fixed by σ against the two it negates, with the factor being their product divided by the fixed square root — 17, 13 and 1 in the three rows.

And here is the counterexample, as a form. Proposition 5.1 says precisely that the quartic form

$$Q(a, b, c, d) = \left(a^2 + 13b^2 - 17c^2 - 221d^2\right)^2 - 52(ab - 17cd)^2$$

represents 25 over every completion $\mathbb{Q}_v$ but represents it over no rational quadruple. It does represent 4, 9 and -25. That is the Hasse principle failing, written out in four variables and degree four — and one sees why nothing like Hasse–Minkowski (Section 5.1) can be expected here: that theorem is about **quadratic** forms.

5.2.2. An Azumaya algebra that registers the obstruction

Equating the norm form to $25e^4$ gives a quartic threefold $X \subset \mathbb{P}^4$ with $X(\mathbb{A}_{\mathbb{Q}}) \neq \varnothing$ and $X(\mathbb{Q}) = \varnothing$ — the points at infinity are excluded too, since a norm form of a **field** is anisotropic, so $e = 0$ forces $a = b = c = d = 0$. Is the failure Brauer–Manin? It must be:

Sansuc. For a torsor under a connected linear algebraic group over a number field, the Brauer–Manin obstruction to the Hasse principle is the only one. The smooth affine $V : N_{L/\mathbb{Q}}(x) = 25$ is a torsor under the norm-one torus $T = R^1_{L/\mathbb{Q}}\mathbb{G}_m$ (if x_0 is one solution, every other is $x_0 t$ with $N(t) = 1$), so **some** class must obstruct.

It can be written down, and it is a quaternion algebra in the two functions already in hand.

Step 1: V fibres over a conic. The map $x \mapsto N_{L/k}(x) = A + B\sqrt{13}$, $k = \mathbb{Q}(\texttt{sqrt 13})$, sends V to the affine conic

$$W : A^2 - 13B^2 = 25, \quad A = a^2 + 13b^2 - 17c^2 - 221d^2, \quad B = 2(ab - 17cd),$$

the very A and B of Section 5.2.1. The fibre over a point of W is the set of x with prescribed $N_{L/k}(x)$ — non-empty exactly when $A + B\sqrt{13}$ is a norm from $L = k\big(\sqrt{17}\big)$.

Step 2: parametrise W. It has the rational point $P_0 = (5, 0)$, so lines $B = m(A - 5)$ through P_0 parametrise it:

$$A = \frac{5(1 + 13m^2)}{13m^2 - 1}, \quad B = \frac{10m}{13m^2 - 1}, \quad A - 5 = \frac{10}{13m^2 - 1}.$$

Modulo squares $A - 5 \equiv 10(13m^2 - 1) = 2f(m)$ with $f(m) = -5(1 - 13m^2)$; and $(17, 2) = 0$ over $\mathbb{Q}$ (as $17 \equiv 1 \bmod 8$ is a square in $\mathbb{Q}_2$ and $2 \equiv 6^2 \bmod 17$), so $(17, A - 5) = (17, f(m))$.

The class.

$$\mathcal{A} = (17, A - 5) = \big(17, a^2 + 13b^2 - 17c^2 - 221d^2 - 5\big) \in \mathrm{Br}(V).$$

It is Azumaya. On the affine conic W the function $A - 5$ vanishes only at P_0, and there to order 2: from $(A - 5)(A + 5) = 13B^2$ with $A + 5 \approx 10$ a unit, $A - 5 = 13B^2/(A + 5)$ and B is a local parameter. It has no poles on affine W. So $\mathrm{div}_W(A - 5) = 2P_0$ has all multiplicities even, every residue of $(17, A - 5)$ vanishes, and the class is unramified on W — hence on V by pullback.

Step 3: why it obstructs. Two facts, one for each kind of place.

(a) At v non-split in $k = \mathbb{Q}(\texttt{sqrt 13})$, $\mathrm{inv}_v\mathcal{A}$ is constant — it does not depend on the local point at all. First, 17 is a square in $\mathbb{Q}_v(\texttt{sqrt 13})$: if 17 is not already a square in $\mathbb{Q}_v$, then 13 and 17 are both non-square **units** there, hence equal modulo squares (the unit group modulo squares has order 2 for odd v), so $221 = 13 \cdot 17$ is a square and $\mathbb{Q}_v(\texttt{sqrt 17}) = \mathbb{Q}_v(\texttt{sqrt 13})$. Therefore $(17, z)_{\mathbb{Q}_v(\texttt{sqrt 13})} = 0$ for every z, and by the projection formula

$$\Big(17, N_{k/\mathbb{Q}}(z)\Big)_v = \mathrm{cor}\Big((17, z)_{k_w}\Big) = 0.$$

Since $f(m) = -5 \cdot N_{k/\mathbb{Q}}\big(1 + m\sqrt{13}\big)$, this gives

$$\mathrm{inv}_v\mathcal{A} = \mathrm{inv}_v(17,-5) \quad \text{for every local point.}$$

(b) At v split in k, local solubility forces $\mathrm{inv}_v\mathcal{A} = 0$. A local point above m exists only if $A + B\sqrt{13}$ is a local norm from L, i.e. $(17, f(m))_w = 0$ at both places $w \mid v$ of k; and $k_w = \mathbb{Q}_v$ there, so this says exactly $(17, f(m))_v = +1$.

Now add up. $(17,-5)$ ramifies exactly at 5 and 17. Of these, 5 is **inert** in $\mathbb{Q}$(`sqrt` 13) (as $13 \equiv 3 \bmod 5$ is a non-residue) while 17 **splits** (as $13 \equiv 8^2 \bmod 17$). So the split places contribute 0 by (b), the non-split places contribute $\mathrm{inv}_v(17,-5)$ by (a), and only $v = 5$ survives:

$$\sum_v \mathrm{inv}_v\mathcal{A}(P_v) = \mathrm{inv}_5(17,-5) = \frac{1}{2} \neq 0 \quad \text{for every } (P_v) \in V\big(\mathbb{A}_{\mathbb{Q}}\big).$$

Hence $V\big(\mathbb{A}_{\mathbb{Q}}\big)^{\mathrm{Br}} = \varnothing$: the Brauer–Manin obstruction is **complete**, and it is carried by a single quaternion algebra.

`csa-brauer.gp` checks (a) directly — 93 values of m against every non-split place up to 40, 2139 symbol comparisons, no deviation from $(17,-5)$ — and confirms that the class is genuinely non-constant, its ramification set moving with m:

m	$f(m)$	ramification of $\mathcal{A}$
0	-5	$\{5, 17\}$
1	60	$\{3, 5\}$
3	580	$\{5, 29\}$
1/2	45/4	$\{5, 17\}$

Two things worth noticing. First, 5 lies in **every** ramification set in the table — that is (a) in action, and it is the whole obstruction. Second, the place carrying the Brauer–Manin obstruction is 5, whereas the contradiction in the proof of Proposition 5.1 appeared at 17. There is no conflict: both are shadows of the same group of order 2, seen through different presentations — once as "which square class can y be", once as "where does the algebra ramify".

Cohomological reading. $\mathrm{Br}(V)/\,\mathrm{Br}(\mathbb{Q}) \cong \mathbb{Z}/2$, generated by $\mathcal{A}$; it is dual to the knot group of the previous box, and the identification is Poitou–Tate duality between $\text{Ш}^1(\mathbb{Q}, T)$ and $\text{Ш}^2\big(\mathbb{Q}, \hat{T}\big)$ for the norm-one torus T. Sansuc's theorem is the statement that this duality is exact — that the descent obstruction and the Brauer–Manin obstruction agree for torsors under connected linear groups.

Cohomological reading. The obstruction is the "knot group" $(K^\times \cap N(\mathbb{A}_L^\times))/N(L^\times)$, and Tate computed it: it is dual to the cokernel of $\oplus_v \hat{H}^{-3}(G_v, \mathbb{Z}) \longrightarrow \hat{H}^{-3}(G, \mathbb{Z})$. Now $\hat{H}^{-3}(G, \mathbb{Z}) \cong H_2(G, \mathbb{Z})$ is the Schur multiplier, which is $\mathbb{Z}/2$ for $G = (\mathbb{Z}/2)^2$ and 0 for G cyclic. So the principle holds automatically for cyclic G (Hasse), and for a biquadratic field it fails with obstruction of order exactly 2 precisely when **no** decomposition group is all of G — which is what was checked above by hand. Equivalently, the obstruction is Ш^1 of the norm-one torus $R^1_{L/K}\mathbb{G}_m$.

5.3. Division algebras over a number field

Theorem 3.1 is a **classification**: a class in $\mathrm{Br}(K)$ is exactly a family $(\mathrm{inv}_v) \in \oplus_v \mathbb{Q}/\mathbb{Z}$ with $\sum_v \mathrm{inv}_v = 0$ (with the constraint that $\mathrm{inv}_v \in \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ at real places and 0 at complex ones). Two consequences that are hard to see any other way:

(a) Index equals exponent. Over a number field, the index of a CSA is the lcm of the denominators of its local invariants, which is also its order in $\mathrm{Br}(K)$.

(b) A division algebra ramifies at at least two places, since a single non-zero invariant cannot sum to zero. In particular there is no division algebra over $\mathbb{Q}$ ramified only at ∞, and none ramified only at one prime.

`csa-brauer.gp` builds these to order with PARI's algebra package:

prescribed invariants	degree	index	lcm of denominators
$1/3$ at 7, $2/3$ at 13	3	3	3
$1/2$ at 2, $1/2$ at 3	2	2	2
$1/2$ at $2, 3, 5, 7$	2	2	2
$1/6$ at 7, $5/6$ at 13	6	6	6
$1/2$ at 7, $1/2$ at 13, in degree 6	6	**2**	2

The last row is $M_3(D)$ for D the quaternion algebra ramified at 7 and 13: **the degree is a choice of presentation, the index is not**. This is the classification doing real work — one reads off the division algebra from a finite list of rational numbers.

5.4. Class field theory, and Grunwald–Wang

Because $\mathrm{Br}(L/K) = K^\times/N(L^\times)$ for cyclic L/K, and because Theorem 3.1 computes Br, one obtains the global reciprocity isomorphism $\mathbb{A}_K^\times/K^\times N(\mathbb{A}_L^\times) \cong \mathrm{Gal}(L/K)$ by bookkeeping with invariants. The same bookkeeping answers **existence** questions: given local behaviour prescribed at finitely many places, is there a global extension realising it? Since one may prescribe local invariants freely subject only to $\sum \mathrm{inv}_v = 0$, the answer is essentially yes — with one famous exception.

Grunwald–Wang, and Wang's counterexample. The number 16 is an 8th power in $\mathbb{Q}_p$ for every **odd** p and in $\mathbb{R}$, but not in $\mathbb{Q}_2$, and not in $\mathbb{Q}$.

Why, explicitly. $x^8 - 16 = (x^2-2)(x^2+2)(x^2-2x+2)(x^2+2x+2)$, so 16 is an 8th power in a field iff one of $2, -2, -1$ is a square there. For odd p the three Legendre symbols $(2/p)$, $(-2/p)$, $(-1/p)$ multiply to $+1$, so at least one of them is $+1$. At $p = 2$ none of $2, -2, -1$ is a square in $\mathbb{Q}_2$.

Grunwald's original theorem asserted more than is true, and the correction (Wang, 1948) is exactly that the prime 2 misbehaves: the "special case" occurs for $8 \mid n$ and involves the failure of $\mathbb{Q}_2^\times$ to be generated in the expected way. `csa-brauer.gp` prints the Legendre symbols place by place. It is a good reminder that the sequence of Theorem 3.1 is **exact** but its consequences for n-th powers are not automatic.

5.5. The Brauer–Manin obstruction

The single most important geometric application, and the one these notes live on. Let X be a smooth variety over K and $\mathrm{Br}(X) = H^2_{\text{ét}(X,\mathbb{G}_m)}$ its Brauer group — concretely, classes of **Azumaya algebras** over X, which one may think of as families of CSAs varying algebraically over X. Evaluating at a point gives, for each $\mathcal{A} \in \mathrm{Br}(X)$,

$$X(K_v) \longrightarrow \mathrm{Br}(K_v) \xrightarrow{\mathrm{inv}_v} \mathbb{Q}/\mathbb{Z}, \quad P \mapsto \mathrm{inv}_v(\mathcal{A}(P)).$$

The obstruction. For $P \in X(K)$, the class $\mathcal{A}(P) \in \mathrm{Br}(K)$ has $\sum_v \mathrm{inv}_v(\mathcal{A}(P)) = 0$ by reciprocity (Theorem 3.1(b)). So

$$X(K) \subseteq X(\mathbb{A}_K)^{\mathrm{Br}} = \left\{ (P_v) \in X(\mathbb{A}_K) : \sum_v \mathrm{inv}_v(\mathcal{A}(P_v)) = 0 \ \text{ for all } \ \mathcal{A} \in \mathrm{Br}(X) \right\},$$

and if $X(\mathbb{A}_K) \neq \varnothing$ but $X(\mathbb{A}_K)^{\mathrm{Br}} = \varnothing$, the Hasse principle fails for a reason one can **compute**.

This is the same product formula as in Section 4, applied fibrewise. In the level-2 case an Azumaya class is literally a quaternion algebra over the function field — which is what `azumaya.gp` in these notes computes, checking that $\mathcal{A} = \left(\prod_{k \neq i}(x - e_k), \prod_{l \neq j}(t - e_l)\right)$ has local invariants matching a twisted Tate pairing, place by place, and that they satisfy reciprocity. `level3.gp` does the degree-3 analogue with cubic symbols, tame and wild.

Cohomological reading. $\mathrm{Br}(X) = H^2_{\text{ét}(X, \mathbb{G}_m)}$; the filtration $\mathrm{Br}_0 \subseteq \mathrm{Br}_1 \subseteq \mathrm{Br}$ (constant, algebraic, transcendental) comes from the Hochschild–Serre spectral sequence, with $\mathrm{Br}_1(X)/\mathrm{Br}_0(X) \cong H^1\left(K, \mathrm{Pic}\left(\overline{X}\right)\right)$. The evaluation pairing is cup product followed by inv_v, and the reciprocity constraint is the exactness of Theorem 3.1 once more.

5.6. Two further homes

Automorphic forms. Quaternion algebras over $\mathbb{Q}$ ramified at a finite even set of places give Shimura curves (indefinite case) or finite sets of ideal classes (definite case), and the Jacquet–Langlands correspondence transfers automorphic forms between GL_2 and the unit groups of these algebras. Eichler's basis problem, the arithmetic of maximal orders and Brandt matrices are all explicit quaternion algebra computations.

Endomorphism algebras of abelian varieties. $\mathrm{End}^0(A) = \mathrm{End}(A) \otimes \mathbb{Q}$ is a semisimple algebra with positive involution, and Albert's classification — a CSA classification — says exactly which algebras occur. Quaternionic multiplication on abelian surfaces is the case where End^0 is an indefinite quaternion algebra over $\mathbb{Q}$; this is precisely the setting in which the Kummer surfaces of these notes acquire extra structure.

6. Crossed products

Everything so far has been built from quaternion and cyclic algebras. Crossed products are the general construction those two are special cases of — and the point of this section is that they are not a further generalisation one **chooses** to make: by Skolem–Noether, **every** central simple algebra with a Galois maximal subfield already is one.

6.1. They are not invented, they are found

Skolem–Noether. Every K-algebra automorphism of a central simple algebra A over K is inner; more generally any two embeddings of a simple K-algebra into A differ by conjugation by a unit of A.

Let A be a CSA of degree n over K and let $L \subset A$ be a maximal subfield with L/K Galois of group G, $|G| = n$. Watch the crossed product appear.

Proposition 6.1. $A = \oplus_{\sigma \in G} L u_\sigma$ for units $u_\sigma \in A^\times$ satisfying

$$u_\sigma x u_\sigma^{-1} = \sigma(x) \quad (x \in L), \qquad u_\sigma u_\tau = c(\sigma, \tau)\, u_{\sigma\tau}$$

for a function $c : G \times G \to L^\times$.

Proof. Each $\sigma \in G$ is a K-embedding $L \to A$, so by Skolem–Noether it is conjugation by some $u_\sigma \in A^\times$. Both $u_\sigma u_\tau$ and $u_{\sigma\tau}$ induce $\sigma\tau$ on L, so $u_\sigma u_\tau u_{\sigma\tau}^{-1}$ centralises L; and by the double centraliser theorem the centraliser of a maximal subfield is itself, $C_A(L) = L$. Hence $u_\sigma u_\tau u_{\sigma\tau}^{-1} \in L^\times$, which defines

c. Finally the u_σ are left L-independent (Dedekind's independence of characters), and n of them each contributing $\dim_K L = n$ gives $n^2 = \dim_K A$, so they span. ■

So the multiplication table of A is completely determined by **one function c on $G \times G$ with values in $L^\times$**. The rest of the theory is the bookkeeping of which c occur and when two of them give the same algebra.

6.2. The definition, and why associativity is the cocycle condition

Turn Proposition 6.1 around into a construction. Given L/K Galois with group G and any $c : G \times G \to L^\times$, set

$$(L/K, G, c) = \oplus_{\sigma \in G} L u_\sigma, \quad u_\sigma x = \sigma(x) u_\sigma, \quad u_\sigma u_\tau = c(\sigma, \tau) u_{\sigma\tau}.$$

When is this associative? Compute both bracketings of $u_\sigma u_\tau u_\rho$:

$$(u_\sigma u_\tau) u_\rho = c(\sigma, \tau) u_{\sigma\tau} u_\rho = c(\sigma, \tau)\, c(\sigma\tau, \rho)\, u_{\sigma\tau\rho},$$

$$u_\sigma \big(u_\tau u_\rho\big) = u_\sigma\, c(\tau, \rho)\, u_{\tau\rho} = \sigma(c(\tau, \rho))\, u_\sigma u_{\tau\rho} = \sigma(c(\tau, \rho))\, c(\sigma, \tau\rho)\, u_{\sigma\tau\rho}.$$

Equating:

$$c(\sigma, \tau)\, c(\sigma\tau, \rho) = \sigma(c(\tau, \rho))\, c(\sigma, \tau\rho).$$

This is exactly the 2-cocycle identity, and it arose from nothing but associativity. A c satisfying it is classically called a **factor set**.

One may normalise $c(1, \sigma) = c(\sigma, 1) = 1$ by rescaling u_1. The resulting algebra is a CSA of degree n over K, split by L.

The split algebra is the trivial factor set. Taking $c \equiv 1$ gives the twisted group algebra $L \rtimes G$, and the map sending $x \in L$ to multiplication by x and u_σ to σ is an isomorphism $(L/K, G, 1) \cong \mathrm{End}_K(L) \cong M_n(K)$. So **split** means **trivial cocycle**, on the nose.

6.3. Changing the u_σ is changing c by a coboundary

The u_σ of Proposition 6.1 were only determined up to $L^\times$. Replacing u_σ by $b_\sigma u_\sigma$ with $b_\sigma \in L^\times$ replaces c by

$$c'(\sigma, \tau) = \sigma(b_\tau)\, b_\sigma\, b_{\sigma\tau}^{-1}\, c(\sigma, \tau),$$

that is, by c times a **coboundary**. So the algebra depends only on the class of c. Conversely two factor sets differing by a coboundary give isomorphic algebras, by the same substitution. And tensoring algebras multiplies factor sets. Hence:

Theorem 6.2 (the crossed product theorem). $c \mapsto (L/K, G, c)$ induces a group isomorphism

$$H^2(G, L^\times) \cong \mathrm{Br}(L/K).$$

Passing to the limit over all finite Galois L/K — legitimate because every CSA has a separable splitting field, hence a Galois one — gives

$$\mathrm{Br}(K) \cong H^2\big(G_K, \overline{K}^\times\big).$$

That is the identification asserted in Section 1, now **derived** rather than quoted, and derived from an explicit multiplication table.

Cohomological reading: this is Galois descent. A CSA of degree n split by L is a K-form of $M_n(K)$, and forms are classified by $H^1(G, \mathrm{Aut}(M_n(L))) = H^1(G, \mathrm{PGL}_n(L))$. The exact sequence

$$1 \longrightarrow L^\times \longrightarrow \mathrm{GL}_n(L) \longrightarrow \mathrm{PGL}_n(L) \longrightarrow 1$$

has $H^1(G, \mathrm{GL}_n(L)) = 1$ (Hilbert 90 in its matrix form, Speiser's theorem), so the connecting map $H^1(G, \mathrm{PGL}_n(L)) \hookrightarrow H^2(G, L^\times)$ is injective — and **the connecting map is precisely "write down the factor set"**. A crossed product is a descent datum with its multiplication table written out.

6.4. The cyclic case: Section 1.2, recovered

Let $G = \langle\sigma\rangle$ be cyclic of order n and $b \in K^\times$. Define

$$c(\sigma^i, \sigma^j) = b^{e(i,j)}, \quad e(i,j) = \begin{cases} 1 \text{ if } i + j \geq n \\ 0 \text{ otherwise,} \end{cases} \quad 0 \leq i, j < n.$$

Because b lies in $K^\times$, σ acts trivially on it and the cocycle identity collapses to an identity between exponents,

$$e(i,j) + e(i+j,k) = e(j,k) + e(i,j+k) \quad \text{(indices mod } n\text{)},$$

which `csa-brauer.gp` verifies for $n = 2, \ldots, 8$: 1295 triples, no failures. Setting $u = u_\sigma$ one gets $u^n = b$ and

$$(L/K, G, c) = (L/K, \sigma, b),$$

the cyclic algebra of Section 1.2. Changing b_σ multiplies b by a norm $N_{L/K}(x)$, so Theorem 6.2 specialises to

$$H^2(\mathbb{Z}/n, L^\times) \cong K^\times / N_{L/K}(L^\times),$$

which is Proposition 1.2 — now proved. Two instances:

- $n = 2$, $L = K(\mathtt{sqrt}\ a)$: the factor set is $c(\sigma, \sigma) = b$ and all else 1, and the algebra is the quaternion algebra (a, b) of Section 1.1.
- $K = \mathbb{R}$, $L = \mathbb{C}$, $\sigma =$ complex conjugation, $b = -1$: the crossed product is **Hamilton's quaternions**, and $\mathrm{Br}(\mathbb{R}) = \mathbb{R}^\times / N(\mathbb{C}^\times) = \mathbb{R}^\times / \mathbb{R}_{>0} = \{\pm 1\}$ — the invariant $1/2$ recorded in Section 2.

`csa-brauer.gp` builds $(\mathbb{Q}(\mathtt{sqrt}\ 5)/\mathbb{Q}, \sigma, b)$ for ten values of b and finds index 1 exactly when b is a norm (no mismatches); confirms that b and $-b$ give the same algebra, since $N(2 + \mathtt{sqrt}\ 5) = -1$ is a coboundary; and confirms that the class of $b = 3$ squares to the class of $9 = N(3)$, hence has exponent 2.

6.5. What the crossed product picture buys

(a) The exponent divides the degree. If A is split by L with $[L : K] = n$, then $\mathrm{res}_{L/K}[A] = 0$ in $\mathrm{Br}(L)$, and since $\mathrm{cor}_{L/K} \circ \mathrm{res}_{L/K}$ is multiplication by n, we get $n[A] = 0$. So the exponent divides the degree — the fact used silently in Section 5.3. (Explicitly, the corestriction is the algebra-theoretic norm; cohomologically it is the transfer.)

(b) Br is a group for a reason. The product of Brauer classes is the product of factor sets; the inverse of c is c^{-1}, realised by the opposite algebra. Without the cocycle description the group law on Br has to be checked by hand on tensor products.

(c) The local invariant is a cyclic crossed product computation. Theorem 2.1 says every class over K_v is $(K_n/K_v, \varphi, \pi^r)$ — a **cyclic** crossed product with L unramified — and inv_v reads off r/n. The whole of Section 2 happens inside one family of factor sets.

6.6. Which algebras are crossed products?

Every Brauer **class** is a crossed product, by Theorem 6.2. But it is a genuinely harder question whether a given division algebra of degree n has a Galois maximal subfield **of degree n** — i.e. is a crossed product "at its own degree" — and harder still whether that subfield can be taken cyclic.

degree of the division algebra	status over a general field
2	always cyclic (a quaternion algebra)
3	always cyclic — Wedderburn (1921)
4	always a crossed product (Albert), with group $\mathbb{Z}/4$ or $(\mathbb{Z}/2)^2$; **not** always cyclic
$p \geq 5$ prime	open whether always cyclic, or even a crossed product
divisible by p^3	**non-crossed products exist** — Amitsur (1972)

Amitsur showed the universal division algebra of degree p^r is not a crossed product for any prime p and any $r \geq 3$; in particular degree 8 already contains non-crossed products. So in general the crossed product description is a statement about Brauer **classes**, not about individual division algebras at their own degree.

Over a number field none of this bites. The Albert–Brauer–Hasse–Noether theorem, in its strong form, says that **every central simple algebra over a number field is a cyclic algebra**: it has a cyclic maximal subfield, of degree equal to its index. One constructs the splitting field directly from the local invariants — a cyclic extension with prescribed local degrees, which exists by the existence theorem of class field theory (and this is one of the places where the Grunwald–Wang correction of Section 5.4 has to be respected).

This is the answer to "how do crossed products connect to the rest of this document". They are the general frame, and over a number field the frame collapses onto its cyclic special case. The quaternion algebras of Section 1.1, the cyclic algebras of Section 1.2, the unramified cyclic algebras that compute inv_v in Section 2, and the algebras with prescribed invariants in Section 5.3 are therefore not a convenient subclass chosen to keep things explicit — **by ABHN they are all of them.** The explicit approach of Section 1 through Section 5 is not a simplification; over number fields it is complete.

Two loose ends worth naming. First, in the geometric setting of Section 5.5 the same picture holds over the function field of a variety: an Azumaya algebra on X is, generically, a crossed product over $K(X)$, and at level 2 it is the quaternion algebra that `azumaya.gp` writes down. Second, the cyclicity of ABHN is exactly what makes the local invariants a **complete** and **computable** invariant: one never has to search for a Galois maximal subfield, because a cyclic one is guaranteed and its degree is the index.

7. Two worked examples

Both are division algebras over $\mathbb{Q}$, given by a basis and a multiplication rule; in each the maximal subfields are computed, and in each the answer turns out to be a classical number-theoretic criterion in disguise.

7.1. Degree 2: Hamilton's quaternions, and the three-square theorem

Take $B = (-1, -1)_{\mathbb{Q}}$, with $\mathbb{Q}$-basis $1, i, j, k$ and

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

— that is $i^2 = j^2 = k^2 = -1$ and $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$. As a crossed product (Section 6) this is $(\mathbb{Q}(i)/\mathbb{Q}, \sigma, -1)$ with σ complex conjugation and $u_\sigma = j$: indeed $jzj^{-1} = \overline{z}$ for $z \in \mathbb{Q}(i)$, and $j^2 = -1$.

The reduced norm is $N(x + yi + zj + wk) = x^2 + y^2 + z^2 + w^2$, anisotropic over $\mathbb{Q}$, so B is a division algebra. Its Hilbert symbols are $(-1,-1)_2 = (-1,-1)_\infty = -1$ and $+1$ elsewhere: **ramified exactly at 2 and ∞**, invariants 1/2 at each, summing to 0 as Section 3 requires.

The quadratic subfields. A **pure** quaternion $v = xi + yj + zk$ has all cross terms cancelling, so

$$v^2 = -(x^2 + y^2 + z^2).$$

Hence $\mathbb{Q}(\sqrt{-m}) \subset B$ if and only if m is a sum of three **rational** squares. Explicitly:

element $v \in B$	v^2	subfield $\mathbb{Q}(v)$
i	-1	$\mathbb{Q}(i)$
$i + j$	-2	$\mathbb{Q}\left(\sqrt{-2}\right)$
$i + j + k$	-3	$\mathbb{Q}\left(\sqrt{-3}\right)$
$j + 2k$	-5	$\mathbb{Q}\left(\sqrt{-5}\right)$
$i + j + 2k$	-6	$\mathbb{Q}\left(\sqrt{-6}\right)$
$j + 3k$	-10	$\mathbb{Q}\left(\sqrt{-10}\right)$

Six pairwise non-isomorphic quadratic fields, the m being distinct and squarefree, all inside one 4-dimensional algebra. And there are infinitely many more.

The bonus: two criteria, one theorem. The Brauer-group criterion of Section 5.3 says $\mathbb{Q}\left(\sqrt{d}\right)$ embeds in B iff it **splits** B, i.e. iff the two ramified places 2 and ∞ are both non-split in $\mathbb{Q}(\texttt{sqrt}\ d)$ — that is, iff $d < 0$ and $d \not\equiv 1 \bmod 8$. Writing $d = -m$ with $m > 0$ squarefree, this reads $m \not\equiv 7 \bmod 8$.

The elementary criterion says instead: m is a sum of three rational squares. By Davenport–Cassels that is the same as a sum of three **integer** squares, and by Gauss and Legendre that happens iff m is not of the form $4^a(8b + 7)$ — for squarefree m, iff $m \not\equiv 7 \bmod 8$.

The two agree, and their agreement is the three-square theorem. `csa-brauer.gp` checks it for all squarefree $m \leq 31$: no disagreements. The first exclusion is $m = 7$:

$$\mathbb{Q}\left(\sqrt{-7}\right) \text{ does not embed in } B,$$

because $-7 \equiv 1 \bmod 8$ makes 2 **split** in $\mathbb{Q}\left(\sqrt{-7}\right)$ — equivalently, because 7 is not a sum of three squares.

And the four-square theorem is next door. The norm form of B is $x^2 + y^2 + z^2 + w^2$, and the Hurwitz order $\mathcal{H} = \mathbb{Z}\langle 1, i, j, (1 + i + j + k)/2\rangle$ — a **maximal** order, unlike the naive $\mathbb{Z}\langle 1, i, j, k\rangle$ — is left- and right-Euclidean for the norm, hence a principal ideal ring. Every prime p therefore has $p = N(h)$ for some $h \in \mathcal{H}$, and clearing the halves gives Lagrange's four-square theorem. The arithmetic of a maximal order in B **is** classical additive number theory.

7.2. Degree 3: a cyclic algebra that sees cubic residues mod 7

Let $\zeta = \zeta_7$ and $\alpha = \zeta + \zeta^{-1}$, so $L = \mathbb{Q}(\alpha) = \mathbb{Q}(\zeta_7)^+$ is the real cyclic cubic field of conductor 7, with

$$f(x) = x^3 + x^2 - 2x - 1, \quad \text{disc } f = 49 = 7^2,$$

the square discriminant confirming that $L/\mathbb{Q}$ is cyclic. Its generator is **explicit**:

$$\sigma(\alpha) = \alpha^2 - 2,$$

because $\alpha^2 - 2 = \zeta^2 + \zeta^{-2}$, i.e. σ is $\zeta \mapsto \zeta^2$. Now put

$$A = (L/\mathbb{Q}, \sigma, 2) = L \oplus Lu \oplus Lu^2,$$

a 9-dimensional $\mathbb{Q}$-algebra with $\mathbb{Q}$-basis

$$1, \alpha, \alpha^2, \quad u, \alpha u, \alpha^2 u, \quad u^2, \alpha u^2, \alpha^2 u^2$$

and the two rules that determine everything:

$$u^3 = 2, \qquad u\,\alpha = (\alpha^2 - 2)\,u.$$

(The second rule moves u past any element of L by applying σ; iterating, $u^2\alpha = \sigma^2(\alpha)u^2$.)

It is a division algebra, and one can see exactly why. At a prime $p \neq 7$ the extension $L_p/\mathbb{Q}_p$ is unramified, so a unit is automatically a local norm and $\mathrm{inv}_p = v_{p(2)}/3$ — non-zero only at $p = 2$, and only because 2 is **inert** in L. At $p = 7$ the extension is totally (tamely) ramified, 7 itself is a norm, and a unit is a local norm exactly when it is a **cube mod 7**; the cubes mod 7 are $\{1, 6\}$, and 2 is not among them. PARI confirms the resulting invariants:

> A has degree 3 and index 3, ramified exactly at 2 and 7, with
>
> $$\mathrm{inv}_2(A) = \frac{1}{3}, \quad \mathrm{inv}_7(A) = \frac{2}{3}, \quad \mathrm{inv}_2 + \mathrm{inv}_7 = 1 \equiv 0 \;\text{ in }\; \mathbb{Q}/\mathbb{Z}.$$

The bonus: the algebra is a cubic residue symbol. For a prime $b \neq 7$,

$$(L/\mathbb{Q}, \sigma, b) \;\text{ is a division algebra } \iff \; b \text{ is not a cube modulo 7.}$$

The reason the two ramified invariants vanish **together** is not luck: p is inert in L iff $p \bmod 7 \notin \{1, 6\}$, and p is a cube mod 7 iff $p \bmod 7 \in \{1, 6\}$ — the same condition. Reciprocity forbids a single non-zero invariant, and here one sees the mechanism.

b	b mod 7	cube mod 7	index	division?
2	2	no	3	yes
3	3	no	3	yes
5	5	no	3	yes
11	4	no	3	yes
13	6	yes	1	no
29	1	yes	1	no
43	1	yes	1	no

The cubic subfields. A cubic field F embeds in A iff it splits A, i.e. iff at each of the two ramified places 2 and 7 there is a **single** prime of F, of local degree 3. Four fields that qualify, and one that does not:

cubic field	disc	$p = 2$: (e, f)	$p = 7$: (e, f)	embeds?
$L = \mathbb{Q}(\zeta_7)^+$	49	$(1, 3)$	$(3, 1)$	yes
$\mathbb{Q}(\zeta_9)^+$	81	$(1, 3)$	$(1, 3)$	yes
$\mathbb{Q}(\sqrt[3]{2})$	-108	$(3, 1)$	$(1, 3)$	yes
$\mathbb{Q}(\sqrt[3]{14})$	-5292	$(3, 1)$	$(3, 1)$	yes
$\mathbb{Q}(\sqrt[3]{5})$	-675	$(1, 1), (1, 2)$	$(1, 3)$	**no**

The discriminants are distinct, so the four are pairwise non-isomorphic. Two of them are cyclic and two are not — so **one division algebra of degree 3 contains both Galois and non-Galois maximal subfields**, which is worth pausing on: the crossed-product description of Section 6 requires a **Galois** maximal subfield, and here it is available (L, or $\mathbb{Q}(\zeta_9)^+$) while $\mathbb{Q}\left(\sqrt[3]{2}\right)$ is equally a maximal subfield and equally useless for that purpose.

Best of all, the generators are visible inside the algebra:

(i) $\alpha \in L \subset A$ generates $L = \mathbb{Q}(\zeta_7)^+$;
(ii) u satisfies $u^3 = 2$, so $\mathbb{Q}(u) \cong \mathbb{Q}\left(\sqrt[3]{2}\right)$;
(iii) $(2-\alpha)u$ has cube $N(2-\alpha) \cdot u^3 = f(2) \cdot 2 = 7 \cdot 2 = 14$, so it generates $\mathbb{Q}\left(\sqrt[3]{14}\right)$.

More generally $(cu)^3 = N_{L/\mathbb{Q}}(c) \cdot 2$ for any $c \in L^\times$, so **every** field $\mathbb{Q}\left(\sqrt[3]{2m}\right)$ with $m \in N_{L/\mathbb{Q}}(L^\times)$ sits inside A — an infinite family of maximal subfields, produced by solving norm equations in a cubic field.

$\mathbb{Q}\left(\sqrt[3]{5}\right)$ is excluded for a reason one can check by hand: $x^3 - 5 \equiv (x+1)(x^2+x+1)$ modulo 2, so 2 splits there, and a prime of local degree 1 cannot support an algebra of local index 3.

8. The dictionary

explicit	cohomological
central simple algebra over K	class in $H^2\left(G_K, \overline{K}^\times\right)$
crossed product $(L/K, G, c)$	a 2-cocycle $c \in Z^2(G, L^\times)$
associativity of the multiplication table	the 2-cocycle identity
rescaling $u_\sigma \mapsto b_\sigma u_\sigma$	changing c by a coboundary
$(L/K, G, 1) \cong M_n(K)$	the trivial class
Skolem–Noether produces the u_σ	the descent datum in $H^1(G, \mathrm{PGL}_n)$
cyclic algebra $(L/K, \sigma, b)$	periodicity: $H^2(\mathbb{Z}/n, L^\times) = K^\times/N(L^\times)$
quaternion algebra (a, b)	cup product in $H^1(K, \mu_2) \times H^1(K, \mu_2)$
Hilbert symbol $(a, b)_v = \pm 1$	inv_v of that cup product
b is a norm from L	the class of b dies in $H^2(\mathrm{Gal}(L/K), L^\times)$
$\mathrm{inv}_v = v(\mathrm{twist})/n$	valuation map $H^2(K_n/K_v, K_n^\times) \to \frac{1}{n}\mathbb{Z}/\mathbb{Z}$
$\prod_v (a, b)_v = 1$	$\sum_v \mathrm{inv}_v = 0$: the global fundamental class
ABHN: split locally $\Rightarrow$ split globally	$Ш^2(K, \mathbb{G}_m) = 0$
Hasse norm theorem for cyclic L/K	$Ш^1\left(K, R^1_{L/K}\mathbb{G}_m\right) = 0$ for cyclic G
its failure for $\mathbb{Q}(\mathtt{sqrt}\ 13, \mathtt{sqrt}\ 17)$	Schur multiplier $H_2((\mathbb{Z}/2)^2, \mathbb{Z}) = \mathbb{Z}/2$
Azumaya algebra on X	class in $H^2_{\mathrm{\acute{e}t}(X, \mathbb{G}_m)}$
Brauer–Manin obstruction	reciprocity applied to the evaluation pairing

9. Two questions about the subfields of A

Both are settled by the one criterion of Section 1.2: since $\mathrm{inv}_2(A) = 1/3$ and $\mathrm{inv}_7(A) = 2/3$ both have order 3, a cubic field L splits A iff every completion of L above 2 and above 7 has local degree 3 — that is, iff **2 and 7 are each non-split in L**, with a single prime above each. And $\deg A = 3$ is prime, so a subfield of A is either $\mathbb{Q}$ or a cubic maximal subfield.

9.1. Which $\mathbb{Q}(\cos(2\pi/m))$ occur

$\left[\mathbb{Q}(\zeta_m)^+ : \mathbb{Q}\right] = \varphi(m)/2$ for $m > 2$, so a **cubic** one needs $\varphi(m) = 6$:

$$\varphi(m) = 6 \iff m \in \{7, 9, 14, 18\},$$

and there are no others — $\varphi(m) \geq \sqrt{m/2}$ bounds the search at $m \leq 72$. But $\mathbb{Q}(\zeta_{14}) = \mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{18}) = \mathbb{Q}(\zeta_9)$, so $m = 14$ and $m = 18$ give nothing new. Both surviving fields do embed:

m	$\mathbb{Q}(\zeta_m)^+$	disc	signature	$\#\{\mathfrak{p} \mid 2\}$	$\#\{\mathfrak{p} \mid 7\}$
$7, 14$	$x^3 - x^2 - 2x + 1$	49	$(3, 0)$	1	1
$9, 18$	$x^3 - 3x - 1$	81	$(3, 0)$	1	1

So the answer is: **there are no others.** The two fields already named exhaust the real cyclotomic subfields of A, for the trivial reason that $\varphi(m) = 6$ has only four solutions and they collapse in pairs. (Degenerately, $\mathbb{Q}(\cos(2\pi/m)) = \mathbb{Q}$ for $m \in \{1, 2, 3, 4, 6\}$, and $\mathbb{Q}$ sits in A for free.)

9.2. Non-real subfields: yes, and two are already above

The guess that A has none is false, and the counterexamples are in Section 7.2′s own list:

field	disc	signature	
$\mathbb{Q}(\zeta_7)^+$	49	$(3, 0)$	totally real
$\mathbb{Q}(\zeta_9)^+$	81	$(3, 0)$	totally real
$\mathbb{Q}\left(\sqrt[3]{2}\right)$	-108	$(1, 1)$	**not** totally real
$\mathbb{Q}\left(\sqrt[3]{14}\right)$	-5292	$(1, 1)$	**not** totally real

$\mathbb{Q}\left(\sqrt[3]{2}\right)$ and $\mathbb{Q}\left(\sqrt[3]{14}\right)$ each have one real place and one complex place. Nothing forbids them, and the reason is worth isolating:

The archimedean place imposes no condition at all. $\mathrm{Br}(\mathbb{R}) = \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ has exponent 2, while A has order 3 in the Brauer group. So $\mathrm{inv}_\infty(A) = 0$ **necessarily** — an algebra of odd degree cannot ramify at a real place — and the splitting criterion there reads $[L_w : \mathbb{R}] \cdot 0 = 0$, which every L satisfies. Only 2 and 7 constrain.

The contrast with Section 7.1 is exact and is probably the source of the intuition. Hamilton's $(-1, -1)$ has $\mathrm{inv}_\infty = 1/2 \neq 0$, so **every** quadratic subfield must be non-split at infinity — which is precisely why they are all **imaginary**, and why the three-square theorem turned up there. Ramification at infinity is impossible in odd degree, so in A that constraint simply evaporates.

Empirically the non-real subfields are, if anything, the more common. Over a box of 2877 distinct cubic fields, 421 embed in A: 187 totally real and 234 with a complex place. The smallest by $|\mathrm{disc}|$ of each kind:

totally real	one complex place
49: $x^3 - x^2 - 2x + 1$	-31: $x^3 + x - 1$
81: $x^3 - 3x - 1$	-76: $x^3 - 2x - 2$
148: $x^3 - x^2 - 3x + 1$	-87: $x^3 - x^2 + 2x + 1$
169: $x^3 - x^2 - 4x - 1$	-108: $x^3 - 2$

The smallest maximal subfield of A of **either** kind is $\mathbb{Q}[x]/\left(x^3 + x - 1\right)$ of discriminant -31 — non-real, and smaller than the totally real record holder $\mathbb{Q}(\zeta_7)^+$.

10. Where this appears in these notes

The survey's chapters 7–10 are an extended computation in this language, and it may help to see which piece is which.

- The **twisted Tate pairing** β_v of the survey is the local invariant inv_v of an explicit quaternion (or degree-ℓ cyclic) algebra; `azumaya.gp` verifies exactly that identification place by place, and checks reciprocity.

- The **tame and wild cubic symbols** of `level3.gp` are the degree-3 Hilbert symbols, i.e. the invariants of cyclic algebras of degree 3, computed by the product formula plus global representatives.
- The observation in `selmer-local-conditions.typ` that "parity is local" and the observation here that "reciprocity is a product formula" are the same fact wearing different clothes: both are $\sum_v \mathrm{inv}_v = 0$.
- The Azumaya algebras of `azumaya.gp` are crossed products over the function field $\mathbb{Q}(X)$: a quaternion algebra is the $G = \mathbb{Z}/2$ case of Section 6, and the pair of descent functions entering it is the factor set.
- `ramification.gp` computes residues of $\mathcal{A}_{ij}$ along the sixteen exceptional curves of $\mathrm{Kum}(E \times E)$ — that is the **ramification** of a Brauer class, the obstruction to it being Azumaya, and it is the geometric analogue of "which places does a division algebra ramify at".

Summary. A central simple algebra over a local field is a valuation and a degree, and nothing more: $\mathrm{inv}_v : \mathrm{Br}(K_v) \cong \mathbb{Q}/\mathbb{Z}$. Over a number field, algebras are classified by their local invariants subject to one condition, $\sum_v \mathrm{inv}_v = 0$, and that condition is reciprocity — quadratic reciprocity when the degree is 2, Artin reciprocity in general, and the Brauer–Manin obstruction when applied fibrewise over a variety. Everything else in this document — Legendre's criterion for conics, the Hasse norm theorem, the classification of division algebras, index = exponent — is a corollary of that one exact sequence, obtained by writing down the right algebra and asking whether it splits.